

4.5 Manipulating Power Series Worksheet

1. Answer each of the following questions

a) Find a series that sums to equal $\ln(1.5)$.

b) Why can't you easily repeat the process from part (a) to find a series that sums to be $\ln(10)$?

c) Find a power series representation for $x \ln(x+1)$.

d) Find a series that sums to $\ln\left(\sqrt[3]{\frac{4}{3}}\right)$.

2. What does the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$ converge to?

3. Find a power series representation for $\sin(x^2)$.

4. Consider the integral $\int_0^1 \cos(x^3) dx$.

a) Find a series which represents this integral.

b) Approximate the value of this integral by calculating the 3rd partial sum of the series.

c) Use a theorem from 3.6 to determine what bound you can place on the error of this approximation.

5. Consider the function $f(x) = \ln(1+x)$.

a) Determine a power series representation for $f(x)$.

b) Use the power series from part(a) to help determine the value of the following series

$$\sum_{k=3}^{\infty} \frac{(-1)^{k+1}}{k} = \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \dots$$

6. Consider the function $f(x) = \frac{1}{1-x}$.

a) Determine a power series representation for $f(x)$.

b) Use the power series from part(a) to help determine the value of the following series

$$\sum_{k=1}^{\infty} \frac{k+1}{2^k} = \frac{2}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \frac{5}{2^4} + \dots$$

(Hint: It will be helpful to manipulate the power series from part(a) using techniques from 6.5)

7. For x in the interval $(-1,1)$, what is the sum of the following series? (Hint: It will be helpful to manipulate the power series for $\frac{1}{1+x}$ using techniques from 6.5)

$$\sum_{n=1}^{\infty} (-1)^{n+1} nx^n = x - 2x^2 + 3x^3 - 4x^4 + 5x^5 - 6x^6 + \dots$$

8. Consider the function $f(x) = \tan^{-1}(x^2)$.

a) Determine a powers series representation for this function and state its interval of convergence.

b) Use part(a) to help show that $\int_0^{1/10} \tan^{-1}(x^2) dx \approx \frac{1}{3000}$.

c) Use a theorem from 3.6 to determine what bound you can place on the error of this approximation.